

Python for Data Science (PDS)

Lab - 13

Roll No. : 128

Name : Riyaz

1. Fetch MCA Faculty Page and Check Status

Send an HTTP GET request to <https://darshan.ac.in/faculty-list/mca> using requests.get().

Display the HTTP status code to verify a successful connection (200) and print the length of the page source code.

```
In [11]: import requests

url="https://darshan.ac.in/faculty-list/mca";

res=requests.get(url);

print(res.status_code)
```

200

2. Extract Web Page Title.

Parse the HTML content using BeautifulSoup with 'html.parser'.

Find and display the content of the title tag to identify the webpage.

```
In [12]: from bs4 import BeautifulSoup as bs

soup=bs(res.text,"html.parser")

print(soup.title.text)
```

Faculty List | MCA | Darshan University, Rajkot

3. Scrape Faculty Names using Heading Tags.

Locate all faculty names displayed on the page (typically enclosed in h3 tag or specific heading tags within faculty cards).

Loop through the results and display the first 5 faculty names cleanly without leading or trailing spaces

```
In [13]: name=soup.select("h2.g-mb-0")

list1=[]

for i in name:
    print(i.text.strip())
    list1.append(i.text.strip())
```

Dr. Gopi Sanghani
Prof. Swati Sharma
Prof. Maulik Trivedi
Prof. Firoz Sherasiya
Prof. Mayur Padia
Prof. Vaseem Ghada
Prof. Umesh Thoriya
Prof. Mehul Bhundiya
Prof. Jayesh Vagadiya
Prof. Rajkumar Gondaliya
Prof. Shruti Maniar
Prof. Dharmik Vasiyani
Prof. Devangi Kotak
Prof. Madhuresh Fichadiya

4. Extract Faculty Designations.

Identify the HTML tag or class used for designations (e.g., Assistant Professor, Associate Professor, Head of Department).

Extract and print the designations of the first 5 faculty members.

```
In [14]: name=soup.select("span.g-mb-10")

list2=[]

for i in name:
    print(i.text.strip())
    list2.append(i.text.strip())
```

[illegible]

5. Scrape Faculty Qualifications.

Find the tags or paragraph elements containing qualification details (such as Ph.D., M.Phil, MCA, M.Tech) for faculty members.

Iterate through them and print the qualification strings.

```
In [15]: name=soup.select("span.g-px-20")

list3=[]

for i in name:
    print(i.text.strip())
    list3.append(i.text.strip())
```

```
Ph.D. , M.E. (CE)
PhD(Pursuing), M.Tech. (CSE)
Ph.D.(Pursuing), M.Tech.(CSE)
Ph.D.(Pursuing), M.E. (CE)
M.E. (CE)
Ph.D. (Pursuing), M.Tech (ICT)
M.Tech. (CSE)
M.E. (CE)
M.E (CE)
M.E. (CE)
ME(CE),BE(IT)
ME(CE)
M. Tech(CS), BE(IT), Diploma I.T.
M.Tech (CE), B.E.(CE)
```

6. Extract Profile Page Links using CSS Selectors

Use `soup.select()` to find anchor tags leading to individual faculty detail profiles.

Extract and print both the faculty member's name (link text) and their profile URL (href).

```
In [16]: name=soup.select("a.g-brd-bottom")

list4=[]

for i in name:
    print(i['href'])
    list4.append("https://darshan.ac.in/"+i['href'])
```

```
/faculty-profile/Dr-Gopi-Bhupendrabhai-Sanghani  
/faculty-profile/Prof-Swati-Rameshbhai-Sharma  
/faculty-profile/Prof-Maulik-D-Trivedi  
/faculty-profile/Prof-Firoz-Ahamadbhai-Sherasiya  
/faculty-profile/Prof-Mayur-Shashikant-Padia  
/faculty-profile/Prof-Vaseem-Jusabbhai-Ghada  
/faculty-profile/Prof-Umeshkumar-Hemubhai-Thoriya  
/faculty-profile/Prof-Mehul-Dilipbhai-Bhundiya  
/faculty-profile/Prof-Jayesh-Dhirubhai-Vagadiya  
/faculty-profile/Prof-Rajkumar-Batukbhai-Gondaliya  
/faculty-profile/Prof-Shruti-Rajeshbhai-Maniar  
/faculty-profile/Prof-Dharmik-Pareshbhai-Vasiyani  
/faculty-profile/Prof-Devangi-Lalitbhai-Kotak  
/faculty-profile/Prof-Madhuresh-Rajeshbhai-Fichadiya
```

7. Extract Faculty Profile Image URLs.

Locate all (img) tags representing faculty photos inside the card containers.

Loop through them and print the image source URL (src) and the alt text attribute for each faculty member.

```
In [17]: name=soup.select("img.img-faculty")  
  
list5=[]  
  
for i in name:  
    print(i['src'])  
    list5.append(i['src'])
```

```
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/5---29-04-2023-1  
1-00-29.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/11---29-04-2023-  
01-44-21.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/8---28-04-2023-0  
2-06-25.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/12---28-04-2023-  
02-06-51.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/13---28-04-2023-  
02-07-48.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/317---28-04-2023  
-01-32-29.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/221---28-04-2023  
-02-08-50.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/294---28-04-2023  
-02-09-43.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/219---28-04-2023  
-02-09-01.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/263---09-05-2023  
-01-10-49.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/336---28-04-2023  
-01-34-37.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/335---27-01-2025  
-09-33-28.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/359---29-04-2023  
-01-45-32.jpg  
https://du-website.s3.ap-south-1.amazonaws.com/U01/Faculty-Photo/331---28-04-2023  
-01-33-39.jpg
```

8. Filter Faculty by Specific Designation or Qualification.

Loop through all scraped faculty designations/qualifications and filter only those holding a specific title (e.g., "Assistant Professor" or containing "Ph.D.").

Store and display the filtered results in a Python list.

```
In [18]: for i in list1:
         if i.startswith("Prof"):
             print(i)
```

```
Prof. Swati Sharma
Prof. Maulik Trivedi
Prof. Firoz Sherasiya
Prof. Mayur Padia
Prof. Vaseem Ghada
Prof. Umesh Thoriya
Prof. Mehul Bhundiya
Prof. Jayesh Vagadiya
Prof. Rajkumar Gondaliya
Prof. Shruti Maniar
Prof. Dharmik Vasiyani
Prof. Devangi Kotak
Prof. Madhuresh Fichadiya
```

9. Extract Faculty Contact / Email Information.

Find all contact links on the faculty cards (such as mailto: email links or phone numbers).

Extract the email addresses or telephone details and display them.

```
In [19]: url=list4[10]
         url="https://darshan.ac.in/"+url
         print(url)

         res=requests.get(url)
         soup1=bs(res.text,"html.parser")

         mon=soup1.select("span.g-font-code")

         for i in mon:
             print(i.text.strip())

         email=soup1.select("span.g-width-80x")

         for i in email:
             print(i.text.strip())
```

```
https://darshan.ac.in/https://darshan.ac.in//faculty-profile/Prof-Shruti-Rajeshbhai-Maniar
```

10. Create a Pandas DataFrame of MCA Faculty Directory.

Collect all faculty Names, Designations, and Profile Links into separate lists.

Combine them into a structured pandas.DataFrame with columns ["Faculty Name", "Designation", "Profile URL"] and display the first 5 rows.

```
In [20]: import pandas as pd

df=pd.DataFrame({'Faculty Name':list1,'Designation':list2,"Profile Image URL":li

print(df)
df.to_csv('output.csv', index=False)
```

	Faculty Name	Designation \
0	Dr. Gopi Sanghani	Dean - Faculty of Computer Application
1	Prof. Swati Sharma	Assistant Professor
2	Prof. Maulik Trivedi	Assistant Professor
3	Prof. Firoz Sherasiya	Assistant Professor
4	Prof. Mayur Padia	Assistant Professor
5	Prof. Vaseem Ghada	Assistant Professor
6	Prof. Umesh Thoriya	Assistant Professor
7	Prof. Mehul Bhundiya	Assistant Professor
8	Prof. Jayesh Vagadiya	Assistant Professor
9	Prof. Rajkumar Gondaliya	Assistant Professor
10	Prof. Shruti Maniar	Assistant Professor
11	Prof. Dharmik Vasiyani	Assistant Professor
12	Prof. Devangi Kotak	Assistant Professor
13	Prof. Madhuresh Fichadiya	Assistant Professor

	Profile Image URL
0	https://darshan.ac.in/faculty-profile/Dr-Gopi...
1	https://darshan.ac.in/faculty-profile/Prof-Sw...
2	https://darshan.ac.in/faculty-profile/Prof-Ma...
3	https://darshan.ac.in/faculty-profile/Prof-Fi...
4	https://darshan.ac.in/faculty-profile/Prof-Ma...
5	https://darshan.ac.in/faculty-profile/Prof-Va...
6	https://darshan.ac.in/faculty-profile/Prof-Um...
7	https://darshan.ac.in/faculty-profile/Prof-Me...
8	https://darshan.ac.in/faculty-profile/Prof-Ja...
9	https://darshan.ac.in/faculty-profile/Prof-Ra...
10	https://darshan.ac.in/faculty-profile/Prof-Sh...
11	https://darshan.ac.in/faculty-profile/Prof-Dh...
12	https://darshan.ac.in/faculty-profile/Prof-De...
13	https://darshan.ac.in/faculty-profile/Prof-Ma...

In []: